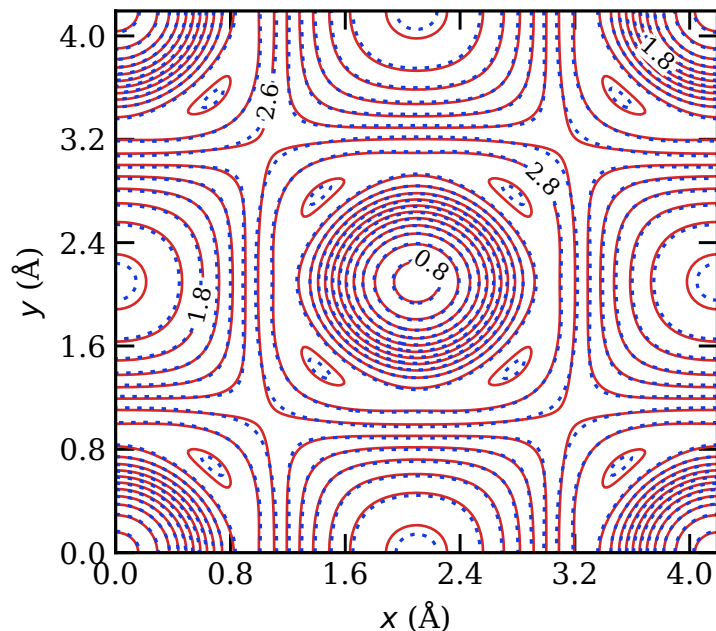


Electron-density contours

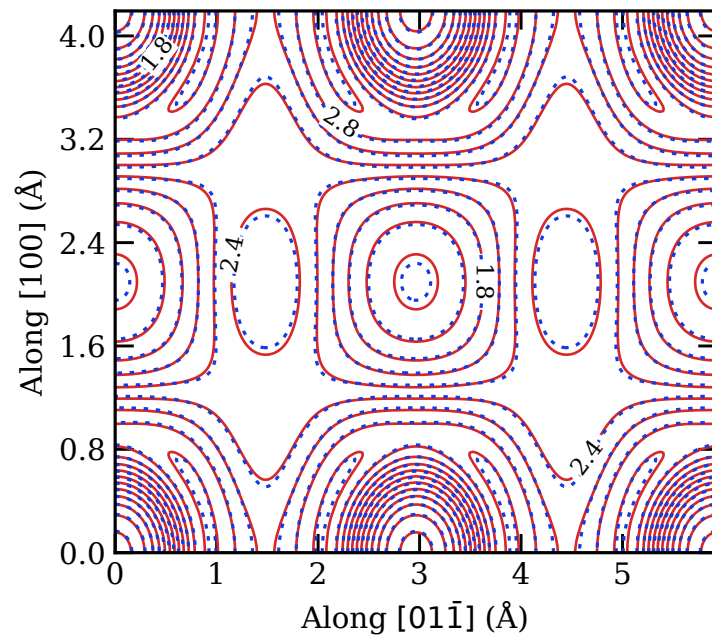
WGC99 (second-order, $\gamma = 2.7$) versus KS · OEPP · PZ-LDA

— GradSCF WGC99 ···· PySCF KS (Gaussian basis)

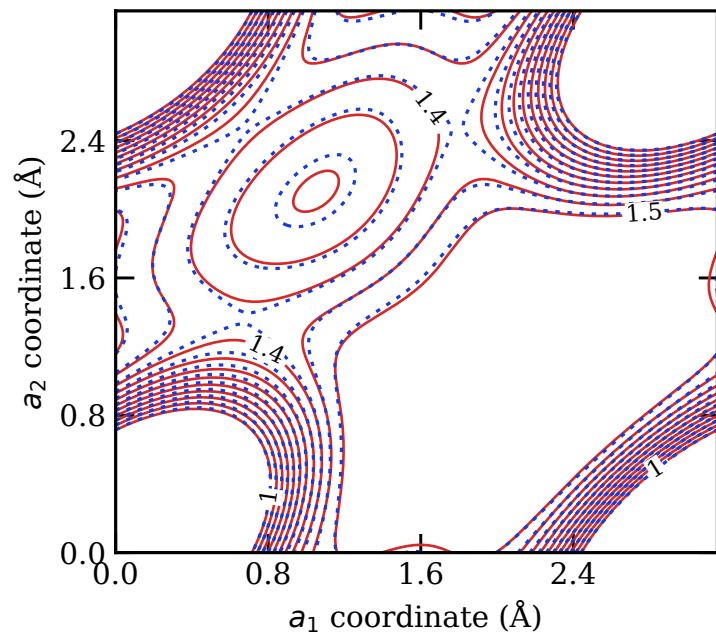
(a) Al (001)



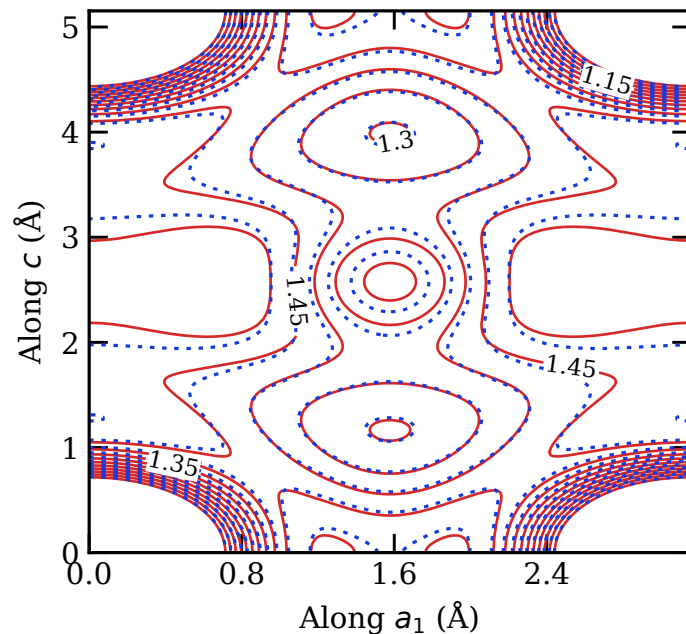
(b) Al (011)



(c) Mg (0001)



(d) Mg (01 $\bar{1}$ 0)



Contour labels show $100n$, with n in electron/Bohr³; coordinates are in Å.
The same contour levels are used for both codes and both sections of each material.
Periodic Fourier interpolation adds sampling points without blurring the calculated density.

Panels have independent axis scales. Mg basal coordinates follow a_1, a_2 (120° apart); ideal c/a is assumed.
KS: Al 10^3 / Mg 8^3 k points, $\sigma = 0.005$ Ha. Finite-basis reference; not original CASTEP data.